

6 (a) at least six horizontal lines equally spaced and arrow to the right B1 [1]

(b) charge used $2e$ C1

$$\text{gain in KE} = 15 \times 1.6 \times 10^{-19} \times 10^3 = 2 \times 1.6 \times 10^{-19} \times V \text{ (p.d. across plates)}$$

or

$$F (= W/d) = 15 \times 1.6 \times 10^{-19} \times 10^3 / 16 \times 10^{-3} \quad \text{C1}$$

$$\text{(hence } V = 7500 \text{ V} \quad \text{or} \quad F = 1.5 \times 10^{-13} \text{ N)}$$

$$E = V/d \quad \text{or} \quad E = F/Q \quad \text{C1}$$

$$E = (7500 / 16 \times 10^{-3}) \quad \text{or} \quad E = (1.5 \times 10^{-13} / 3.2 \times 10^{-19})$$

$$E = 4.7 \times 10^5 \text{ (468 750) V m}^{-1} \quad \text{A1 [4]}$$

or

$$\text{KE} (= \frac{1}{2}mv^2) = 15 \times 10^3 \times 1.6 \times 10^{-19}$$

$$v = [(2 \times 15 \times 10^3 \times 1.6 \times 10^{-19}) / (6.68 \times 10^{-27})]^{1/2} = 8.5 \times 10^5 \text{ m s}^{-1} \quad \text{(C1)}$$

$$a (= v^2/2s) = (8.5 \times 10^5)^2 / 2 \times 16 \times 10^{-3} = 2.25 \times 10^{13} \text{ m s}^{-2}$$

$$F (= 6.68 \times 10^{-27} \times 2.25 \times 10^{13}) = 1.5 \times 10^{-13} \text{ N}$$

$$E = F/Q \quad \text{(C1)}$$

$$Q = 2e \quad \text{(C1)}$$

$$E = 4.7 \times 10^5 \text{ V m}^{-1} \quad \text{(A1)}$$

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